

Chapter 1: Large Numbers Around Us

1. 'X' is a 6-digit number formed using exactly three different digits. One of the digits appears once, another appears twice, and the third appears three times. When 2 lakhs are added to 'X', the resulting number 'Y' is still a 6-digit number. What is the highest possible value of 'Y'?

a) 977889 b) 999999 c) 999988 d) 999887

2. What is the product of all the numbers on a telephone dial-pad?

a) 12345 b) 32451 c) 362880 d) None of these

3. Sam forms two different 6-digit numbers, X and Y, using digits from 0 to 9 (without repetition within a number). Exactly two digits are common between X and Y.

- The digit 9 appears in the same position in both numbers
- Y ends with 0, has only one even digit, and its digits are arranged in descending order
- 6 is present in X, and the number of even digits on its left is equal to the number of even digits on its right

If X is the largest possible number, what is the difference between X and Y?

a) 9310 b) 9210 c) 8420 d) 8310

4. If each of the rows follows the same theme, what will come in place of “?”

$$\boxed{1 \text{ } \bullet \text{ } 8 \text{ } \bullet \text{ } 2 \text{ } \bullet} > \boxed{1 \text{ } \bullet \text{ } \bullet \text{ } 8 \text{ } \bullet \text{ } 2 \text{ } \bullet} \quad \boxed{\bullet \leq 7}$$

$$\boxed{\bullet \text{ } \bullet \text{ } 3 \text{ } 5 \text{ } \bullet \text{ } 7} < \boxed{\bullet \text{ } 3 \text{ } 5 \text{ } \bullet \text{ } \bullet \text{ } 7} \quad \boxed{\bullet \leq 3}$$

$$\boxed{\bullet \text{ } 9 \text{ } 6 \text{ } \bullet \text{ } 3 \text{ } \bullet} > \boxed{\bullet \text{ } 9 \text{ } 6 \text{ } \bullet \text{ } \bullet \text{ } 3} \quad \boxed{\bullet \leq 2}$$

$$\boxed{\bullet \text{ } \bullet \text{ } 5 \text{ } 0 \text{ } 6 \text{ } \bullet} < \boxed{\bullet \text{ } 5 \text{ } \bullet \text{ } 6 \text{ } 0 \text{ } \bullet} \quad \boxed{?}$$

a) $\boxed{\bullet \leq 6}$

b) $\boxed{\bullet \leq 4}$

c) $\boxed{\bullet \leq 5}$

d) $\boxed{\bullet \leq 9}$

5. What will come in place of “?”

Crore - re	→	00000
Million - ion	→	000
Thousand - d	→	00
Billion - llion	→	0000
Lakh - kh	→	?

a) 0 b) 00 c) 000 d) 0000

- The difference between the first and last digits is as small as possible
- The hundreds digit is double the thousands digit
- No consecutive digits appear next to each other
- All given tokens must be used exactly once

LEFT **2** **5** **8** **1** **4** **7** RIGHT

Number Formed _____

- Page 12



The Thinking Spot

Columns A, B, and C contain 3 numbers each, such that:

- The number of times each number appears in all 3 columns combined, is equal to its value (e.g., 2 appears twice, 3 appears thrice, etc.)
- Each column contains AT LEAST TWO different numbers

If column A is the only column with THREE different numbers, then which column has the highest sum?

A	B	C
○	②	○
○	○	○
④	○	③

- (a) Column A
(c) Column C

- (b) Column B
(d) Both columns B and C

